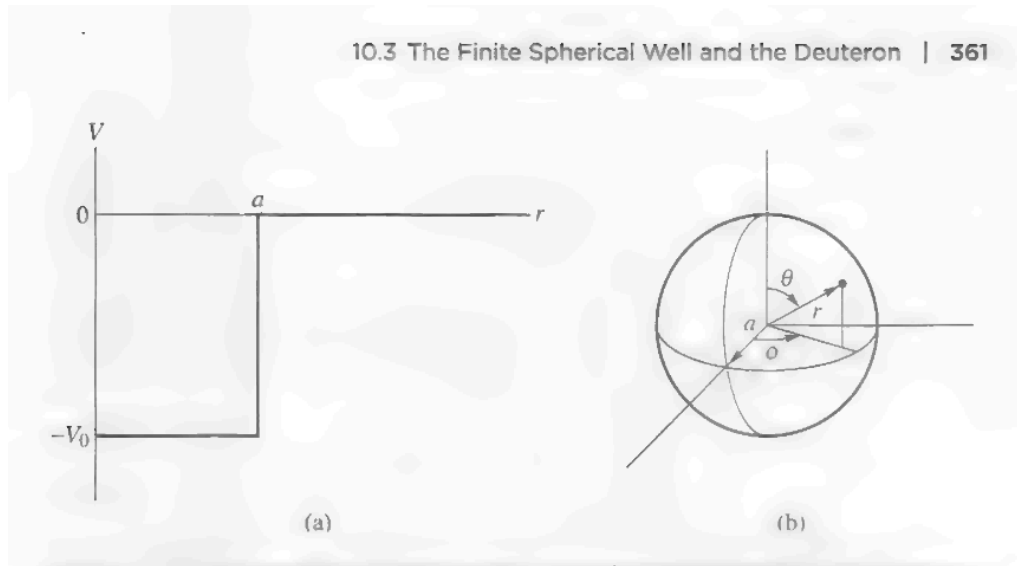


10.3-10.6

Questions for Gallicchio

Can you explain spin orbit coupling again?

10.3 The finite Spherical Well and the Deuteron



Deuteron has intrinsic spin 1. Nuclear force is spin dependent. Spin orbit coupling causes not spherically symmetric stuff?

10.4 The infinite spherical well

Spherical bessel and Neumann functions

Known as the **spherical Bessel equation**. Solutions to this equation that are regular at the origin are called **spherical Bessel functions**,

$$j_l(\rho) = (-\rho)^l \left(\frac{1}{\rho} \frac{d}{d\rho} \right)^l \left(\frac{\sin \rho}{\rho} \right) \quad (10.68a)$$

while irregular solutions at the origin are called **spherical Neumann functions**,

$$\eta_l(\rho) = -(-\rho)^l \left(\frac{1}{\rho} \frac{d}{d\rho} \right)^l \left(\frac{\cos \rho}{\rho} \right) \quad (10.68b)$$

The first few functions, shown in Fig. 10.10, are

$$\begin{aligned} j_0(\rho) &= \frac{\sin \rho}{\rho} & \eta_0(\rho) &= -\frac{\cos \rho}{\rho} \\ j_1(\rho) &= \frac{\sin \rho}{\rho^2} - \frac{\cos \rho}{\rho} & \eta_1(\rho) &= -\frac{\cos \rho}{\rho^2} - \frac{\sin \rho}{\rho} \\ j_2(\rho) &= \left(\frac{3}{\rho^3} - \frac{1}{\rho} \right) \sin \rho - \frac{3 \cos \rho}{\rho^2} & \eta_2(\rho) &= -\left(\frac{3}{\rho^3} - \frac{1}{\rho} \right) \cos \rho - \frac{3 \sin \rho}{\rho^2} \end{aligned} \quad (10.69)$$

$$E_{n_r=1, l=1} = \frac{\hbar^2 k^2}{2\mu} = \frac{\hbar^2}{2\mu} \left(\frac{4.49}{a} \right)^2$$

$$\begin{aligned} & \left[-\frac{1}{X} \frac{\hbar^2}{2\mu} \frac{d^2 X}{dx^2} + \frac{1}{2} \mu \omega^2 x^2 \right] + \left[-\frac{1}{Y} \frac{\hbar^2}{2\mu} \frac{d^2 Y}{dy^2} + \frac{1}{2} \mu \omega^2 y^2 \right] \\ & + \left[-\frac{1}{Z} \frac{\hbar^2}{2\mu} \frac{d^2 Z}{dz^2} + \frac{1}{2} \mu \omega^2 z^2 \right] = E \end{aligned} \quad (10.85)$$

$$-\frac{\hbar^2}{2\mu} \frac{d^2 X}{dx^2} + \frac{1}{2} \mu \omega^2 x^2 X = E_x X \quad (10.86a)$$

$$-\frac{\hbar^2}{2\mu} \frac{d^2 Y}{dy^2} + \frac{1}{2} \mu \omega^2 y^2 Y = E_y Y \quad (10.86b)$$

$$-\frac{\hbar^2}{2\mu} \frac{d^2 Z}{dz^2} + \frac{1}{2} \mu \omega^2 z^2 Z = E_z Z \quad (10.86c)$$

$$E_n = \left(n + \frac{3}{2}\right) \hbar \omega \quad n = 2n_r + l \quad n = 0, 1, 2, \dots \quad (10.96)$$

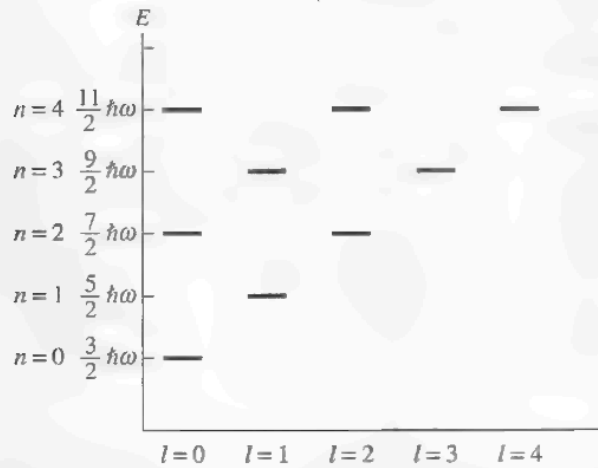


Figure 10.14 The energy levels $E_n = (n + \frac{3}{2})\hbar\omega$ of the isotropic harmonic oscillator, showing the degeneracy.

10.6

parabolic coordinate systems?